higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 6 pages.

Copyright reserved

QUESTION 1: GENERAL

1.1
1.1.1 False
1.1.2 True
1.1.3 False
1.1.4 True
1.1.5 True

(5 × 1) (5)

1.2
1.2.1 B
1.2.2 C
1.2.3 D
1.2.4 B
1.2.5 B

(5 × 1) (5)
[10]

QUESTION 2: INTRODUCTION AND BACKGROUND

2.1
- Vertical-twist tie
- Butterfly wall ties
- Modified-PWD wall tie
- Single-wire wall tie
- Double-triangle wall tie
- Crimp-wire tie

(Any 3 × 1) (3)

2.2 The overall thickness of the cavity wall will be 270 mm. (1)

2.3
2.3.1 Hollow masonry units contain a core that is more than 25% of its gross volume.✓ Hollow units are most commonly used in structural applications. Fewer materials are used and they are lighter which makes it easy to work with.✓

2.3.2 Solid masonry units contain no core or if it has a core it is not bigger than 25% of its gross volume.✓ Solid units are used in composite masonry or veneers.✓

(2 × 2) (4)
[8]

QUESTION 3: SETTING OUT MASONRY WALLS

3.1
- This method is based on the theorem of Pythagoras.
- When using this method a steel peg is put at point A and three units are measured to point B.
- From point B four units are measured to point C.
- If point A to point C is measured and it equals five units then the corner at point B is a right angle or squared.
- If it does not equal five units then the corner is not at a right angle and it must be checked again. (Any relevant answer) (5)

3.2 The position of the walls must be marked on the foundation. Profiles must be positioned at the marked ends of the walls to be built.✓ Apply the stays✓ to the profile and use a spirit level to plumb the profile.✓ When the profile is plumb lock the stays by using heavy objects or blocks.✓ Once the profile is plumb and stayed correctly use the gauge rod to transfer marks of equal gauge onto the profile.✓ (Any relevant answer) (5)

3.3 3.3.1
- Screed rails act as a guide when placing concrete.
- Screed rails hold the concrete while it sets.

3.3.2
- When screed rails are removed after use a wire brush should be used to remove all hardened concrete stuck to the screed rails.
- After removing the hardened concrete the screed rails should be washed with water and dried before being stored.

(2 × 2) (4)

3.4
3.4.1 Joint ruler or raking tool
3.4.2 Steel square
3.4.3 Bricklayer's trowel
3.4.4 Builder's line
3.4.5 Plaster comb
3.4.6 Chalk line

(6 × 1) (6)

3.5
- The allowance made for the variance in the height of the mortar will vary from site to site.
- Factors such as the height of windows, doors or other architectural features will influence the thickness of the mortar.
- The size of the windows or doors might also vary as well as the size of masonry units to be built with. (Any relevant 3 × 1) (3)

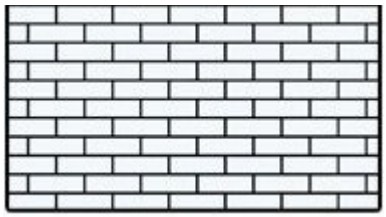
[23]

QUESTION 4: BASIC CONSTRUCTION OF MASONRY WALLS

4.1 Marking rubric per bond. Accept relevant alternate drawings.

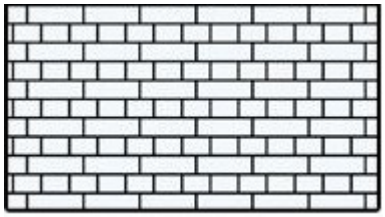
Correct bonding	2 marks
Type of bond	1 mark
Neatness	1 mark
TOTAL:	**4 marks**

4.1.1



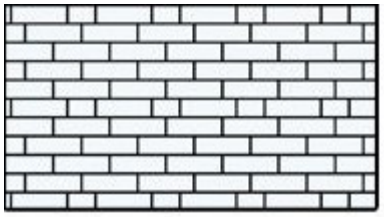
Stretcher bond

4.1.2



English bond

4.1.3



Flemish bond

(3 × 4) (12)

4.2 4.2.1 Horizontal DPC should be laid at plinth level.✓ Lay the DPC to the full thickness of the wall.✓ The DPC should be lapped at least 150 mm at all joints.✓ At ends the DPC must overlap the full width of the wall.✓ All horizontal DPC must protrude at least 5 mm from the external face of the wall.✓ The DPC should not be carried across openings and doorways.✓ (Any relevant answer) (6)

4.2.2
- Some types of DPC do not bond with mortar.
- The stability of the wall will be compromised.
- There will be unnecessary extra cost and time because the wall is not structural. (Any 3 × 1) (3)

4.3 Clay and concrete masonry units have different movement patterns,✓ they respond differently to temperature, moisture and stress✓ and cracking will occur where they make contact with each other.✓ (Any relevant answer) (3)

4.4
- When they are in separate leaves with no mortar in between and the leaves are joined by wall ties
- When there is a vertical control joint between the clay and concrete units
- When the lower course of a foundation wall is of one material and the upper course is of another material they can be separated by a horizontal slip joint (DPC can be placed in the bedding course at the junction of the two materials) (Any relevant 3 × 1) (3)

4.5 A unit is an individual brick or block. (1)
[28]

QUESTION 5: ADVANCED CONSTRUCTION OF MASONRY WALLS

Accept relevant alternate answers for 5.1.1 to 5.1.7

5.1

5.1.1 A roof anchor is used to tie metal straps to masonry walls or concrete.✓ Roof anchors tie a wall to the roof truss.✓

5.1.2 A brick arch is a structure built over an opening.✓ The arch also supports the weight of the structure. Arches can be built in various straight or curved forms.✓

5.1.3 A keystone is the centre brick of the arch✓ that locks the arch together.✓

5.1.4 A control joint is a groove inserted to control cracking.✓ The control joint helps to prevent the concrete cracking on its own in an uncontrolled manner.✓

5.1.5 A composite beam is a beam consisting of two or more different materials joined together.✓ When the two materials are joined together they are stronger than each material separately.✓

5.1.6 Prestressed concrete lintels are made by putting the steel in the concrete under tensile strain while being manufactured.✓ It can function as a composite beam and utilise the strength and hardness of both materials.✓

5.1.7 A gauge rod is a long straight edge marked at intervals equal to the thickness of the masonry unit plus the mortar joint.✓ It is used for checking and keeping masonry courses even.✓

(7 × 2) (14)

5.2 The lintel must cover the opening by a minimum of 15 mm and a maximum of 30 mm.✓ Create a timber frame on the inside of the opening for the lintel to rest on.✓ Lay mortar over the frame of the door/window and ensure that the wall plate is level with the top of the lintel. Use a spirit level to ensure that the lintel is level.✓ Place DPC over the concrete lintel and put sealant under the lintel to prevent moisture entering.✓ Determine the minimum number of courses and lay each course with brick force in the next three consecutive courses.✓ Ensure that each course is level and plumb. The cavity (if a cavity wall) must be filled with mortar or infill concrete. Wait seven days (or more) and remove the timber frame.✓ (Any relevant answer) (6)

5.3 If lintels are not cured it would not be able to bear a load✓ and would crack.✓ (Any relevant answer) (2)

5.4 Racking back is a method used to join brickwork by creating a step-like structure.✓ It is formed by deducting 110 mm of each course and provides no problem when joining up is required.✓ Toothing is a process used to join new brickwork to existing brickwork by deducting 110 mm on every alternate course✓ and allowing for bricks to be interlocked as the wall construction proceeds.✓ (Any relevant 2 + 2) (4)

5.5
- Cement
- River/Umgeni sand
- Concrete stones
- Lime
- Crushed stone
- Water (Any relevant 4 × 1) (4)

5.6 Roof anchors tie a wall to the roof trusses in a fixed position. (Any relevant answer) (1)

[31]

TOTAL: 100

Copyright reserved